

CONTENTS

57TH YEAR OF PUBLICATION
electronics
 SINCE JANUARY 1969 **FOR YOU**

TECHNOLOGY

26 CONSUMER: Amazing Smart Projectors Can Now Replace Large TVs

36 FUTURE: The Emerging Wonderland Of 'Living' Computer Systems

42 INTERVIEW: "It's Not About How Many Companies You Incubate—It's About How Many Are Still Standing Five Years Later."
 — Dr M.J. Shankar Raman, Chief Executive Officer, IITM Pravartak

48 INNOVATIONS: Magnetic Signals You Couldn't Hear—Until Now

BUSINESS


54 INTERVIEW: "The Czochralski Method Is Preferred For Wafers Over 203mm" — Rajasekar Elavarasan, Founder, Raana Semiconductors Private Limited

70 INTERVIEW: "India's Semiconductor Ecosystem May Surpass China Over Time" — Yutaka Ito, Natalie Abe, and Satoshi Teshima, TANAKA Group

72 PURCHASE: Buyers' Guide: Leading Suppliers Of SMT Selective Soldering

78 PROFITS: Making Electronics Manufacturing Predictably Profitable Using Data Analysis

DESIGN

80 CIRCUIT: 5 Interesting Reference Designs For Sensors

82 GUIDELINES: Why Design For Manufacturing Is India's Electronics Game-Changer

87 PRODUCT: Bulletproofing Your Product Design For Wireless Products


DO-IT-YOURSELF

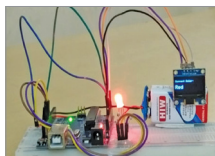
92 PROJECT: Colour Buddy: Learn Colours With Lights

95 CIRCUIT IDEA: Dual-Tone Single Bell For Two Doors

97 PROJECT: Smallest Portable Game Boy And Wi-Fi Hacking Device

102 PROJECT: Real-Time Temperature And Humidity Monitoring System Using TFT Display

104 PROJECT: AI Vision Assistant Camera

107 PROJECT: Compact AI Health Monitoring And Anomaly Detection Device


REGULARS

8 Research & Innovation Updates
16 Truly Innovative Electronics
20 New Products
52 Dashboard: Key Industry Trends

 That Can Shape The Future Of
 Your Business

56 Innovators & Startups
64 Industry News: Industry Updates

EDITOR : Rahul Chopra

EDITORIAL CORRESPONDENCE : Phone: +91-9811155335
 E-mail: editor@efy.in

SUBSCRIPTIONS & MISSING ISSUES : Phone: +91-9811155335
 E-mail: support@efy.in

BACK ISSUES, BOOKS, KITS, COMPONENTS, etc : Phone: +91-9811155335
 E-mail: support@efy.in

NEWSSTAND DISTRIBUTION : Phone: +91-9811155335
 E-mail: efycirc@efy.in

ADVERTISEMENTS
NEW DELHI : Phone: +91-9811155335
 (HEAD OFFICE) E-mail: efyenq@efy.in

MUMBAI : E-mail: rmowest@efy.in

BENGALURU : E-mail: rmosouth@efy.in

CHINA : Worldwide Focus Media
 E-mail: china@efy.in

GERMANY : PMS Plantenberg Media Service GmbH
 E-mail: germany@efy.in

JAPAN : Tandem Inc.
 E-mail: japan@efy.in

TAIWAN : J.K. Media
 E-mail: taiwan@efy.in

UNITED KINGDOM : ASA Media
 E-mail: uk@efy.in

UNITED STATES : E & Tech Media
 E-mail: usa@efy.in

Printed, published and owned by Ramesh Chopra. Printed at Thomson Press India Private Limited, 18/35 Delhi Mathura Road, Faridabad 121007, Haryana, on the first day of each month and published from D-87/1, Okhla Industrial Area, Phase-1, New Delhi 110020. Copyright 2025. All rights reserved throughout the world. Reproduction of any material from this magazine in any manner without the written permission of the publisher is prohibited. Although every effort is made to ensure accuracy, no responsibility whatsoever is taken for any loss due to publishing errors. EFY will not be responsible for any wrong claims made by an advertiser. Disputes, if any, will be settled in a New Delhi court only.

SUBSCRIPTION RATES

EFY = Electronics For You

PRINT MAGAZINE	1 YEAR (12 copies each)	3 YEARS (36 copies each)	5 YEARS (60 copies each)
WITHIN INDIA (IN RUPEES)			
EFY (Rs 100/copy)	960	2700	4200
SAARC COUNTRIES (IN US\$ BY AIR)			
EFY	57	155	NA
OTHER COUNTRIES (IN US\$ BY AIR)			
EFY	114	310	NA

Courier delivery within India @ Rs 50 per copy

e-magazine subscriptions within India are available at half the rates mentioned here.

Overseas rates for each e-magazine in US\$:

1 year: \$12; 3 years: \$33; 5 years: \$50 only

 Please send payments only in favour of **EFY Enterprises Pvt Ltd** through demand draft/cheque/money order to D-87/1, Okhla Industrial Area, Phase 1, New Delhi 110020, or online at: <http://subscribe.efyindia.com> or scan the QR code.
